

Troubleshooting and Tool Report

Taylor Myers

Network Problem Selected

The network problem I selected is that users are unable to connect to the globally dispersed servers even when the internet connection is working. This is a serious problem because these servers are used for command-and-control operations, so losing access to them could interfere with the team's ability to complete its mission. Since the servers are located in different areas around the world, the team needs a way to determine where the network traffic is going and where the connection may be failing. Routing plays an important role in determining which parts of a network are reachable and how packets travel to their destination, and a routing disruption can make parts of a network unreachable (Dainotti et al., 2014).

Troubleshooting Steps

The first step is to identify the problem by confirming which global servers cannot be reached and determining whether the issue affects one host or multiple hosts. The second step is to establish a theory of probable causes, such as a routing problem, firewall issue, or failure somewhere along the network path. The third step is to test the theory to determine the cause by using network troubleshooting tools to see where communication is stopping. Next, the administrator should establish a plan of action and identify any possible effects before making changes. The fifth step is to implement the solution or escalate the problem if necessary. After that, the administrator should verify full system functionality and implement preventive measures to make sure access to the global servers has been restored. The final step is to document the findings, actions taken, and results so the information can be used if the problem happens again.

Tool and Description

The tool I would use to troubleshoot this problem is `tracert` on Windows, which is called `traceroute` on Linux. `Tracert` is a network diagnostic tool that shows the path packets take from the local computer to a remote destination. It identifies the routers, or hops, that traffic passes through along the way and can help determine where communication is slowing down or failing. Viger et al. (2008) explain that `traceroute` reports the IP address of each network-layer device along the path from a source to destination and is commonly used to diagnose network problems. This makes it useful for troubleshooting the connection to the globally dispersed servers because it can help show whether the problem is occurring on the local network or farther along the route. Research on Internet outages has also used `traceroute` measurements to examine forward-path information and determine whether destinations were reachable during network disruptions (Dainotti et al., 2014).

Tool Operational Use Case

Tracert can also be used in the future when users are having trouble reaching remote servers, websites, or other network resources. A network administrator can run the command to see the path traffic takes and identify the last successful hop before communication fails. This can help determine whether the issue is within the local network, with an internet service provider, or closer to the destination. Traceroute has also been used in network outage analysis to test reachability and examine how traffic moves through different network paths during disruptions (Dainotti et al., 2014).

Tool Functionality

The basic Windows command is tracert followed by a hostname or IP address, such as tracert 8.8.8.8. In the uCertify lab, the command tracert www.lammle.com showed the route to the destination across 10 hops. The command shows each hop that packets pass through on the way to the destination and displays the response time for each hop. Traceroute works by sending probe packets with increasing Time to Live (TTL) values. Each router decreases the TTL value, and when it reaches one, the router returns an ICMP Time Exceeded message, which allows the tool to identify that hop (Viger et al., 2008). The results also showed the hostname and IP address for the devices along the route. This helps identify where delays or failures may be occurring along the route. Common options include -d, which prevents hostname resolution and can make the trace faster; -h, which sets the maximum number of hops; -w, which sets how long tracert waits for a response; -4, which forces IPv4; and -6, which forces IPv6. These options allow the administrator to adjust the test depending on the network problem being investigated.

```
C:\Users\tlammle>tracert www.lammle.com

Tracing route to lammle.com [206.123.114.186]
over a maximum of 30 hops:
  1    1 ms    <1 ms    <1 ms    dslmodem.domain.actdsitmp [192.168.0.1]
  2   53 ms   52 ms   52 ms   hlrn-dsl-gw36-228.hlrn.qwest.net
[207.225.112.228]
  3   52 ms   53 ms   52 ms   hlrn-agw1.inet.qwest.net [71.217.189.25]
  4   75 ms   75 ms   74 ms   dal-core-01.inet.qwest.net [67.14.2.53]
  5   76 ms   76 ms   76 ms   dap-brdr-01.inet.qwest.net [205.171.225.49]
  6   76 ms   76 ms   76 ms   205.171.1.110
  7   75 ms   76 ms   106 ms  xe-0-0-0.er2.dfw2.us.above.net [64.125.26.206]
  8   76 ms   76 ms   76 ms   209.249.119.74.available.above.net
[209.249.119.74]
  9   76 ms   76 ms   76 ms   65.99.248.250
10   76 ms   76 ms   76 ms   pageuppro.pageuppro.com [206.123.114.186]

Trace complete.
```

References

- Dainotti, Alberto, Squarcella, Claudio, Aben, Emile, Claffy, Kimberly C., Chiesa, Marco, Russo, Michele, & Pescapé, Antonio. (2014). Analysis of Country-Wide Internet Outages Caused by Censorship. *IEEE/ACM Transactions on Networking*, 22(6), 1964–1977.
<https://doi.org/10.1109/TNET.2013.2291244>
- Viger, Fabien, Augustin, Brice, Cuvellier, Xavier, Magnien, Clémence, Latapy, Matthieu, Friedman, Timur, & Teixeira, Renata. (2008). Detection, understanding, and prevention of traceroute measurement artifacts. *Computer Networks (Amsterdam, Netherlands : 1999)*, 52(5), 998–1018.
<https://doi.org/10.1016/j.comnet.2007.11.017>